Do Now · retrieval

Binary search: trace and code

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Does linear search require ordered input?

Q2

What must be true of the two input runs when merging them?

Q3

For seven elements, what is the last zero-based index?

Q4

Find the integer midpoint of low=2 and high=8.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Sort before searching — Binary search relies on a sorted array; otherwise discarding half may lose the target.

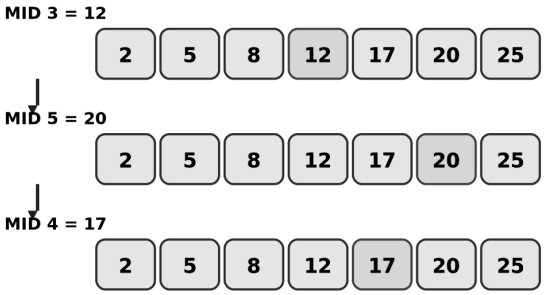
Core idea 2

Inspect the middle — Compare target to the midpoint and keep only the half that could contain it.

Core idea 3

Stop correctly — Return an index when found; return -1 if low becomes greater than high.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

precondition — Input assumption: the array is ordered.

low — Lower inclusive search index.

high — Upper inclusive search index.

mid — Index of the middle candidate.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

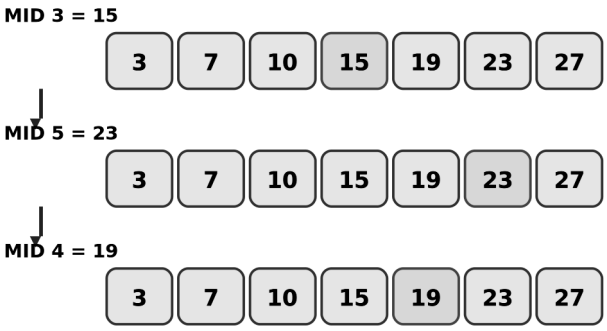
Search for 19 in [3,7,10,15,19,23,27]. Record every midpoint checked.

Follow the worked method

1. Predict first — Use inclusive low and high indexes.

2. Model solution — mid 3 → 15, so low=4; mid 5 → 23, so high=4; mid 4 → 19. Return 4.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Which midpoint is checked first?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Search for 8 in [2,5,8,11,14]. List midpoint indexes.

Question 2

Search for 13 in [2,4,6,8,10,12,14]. List indexes checked and final return.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why setting low = mid instead of mid + 1 can cause failure.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Binary search is always safe on an unsorted array.

Use key vocabulary from memory

1. Explain precondition in your own words.

2. Explain low in your own words.

3. Explain high in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.